Indian Institute of Technology Jodhpur
MAL1020
Tutorial Sheet 9

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1. Solve by the method of variation of parameters:

$$x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} = 1, \quad x > 0$$

It being given that $y = x^{-1}$, $y = 1$, $y = x$ are three linearly independent solutions of its reduced equation.

2. Solve the system of differential equations:

$$\frac{d}{dt}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

3. Solve the differential equation using the power series method about $x = 0$:

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

4. Solve the differential equation using the power series method about $x = 1$:

$$(x - 1)y'' + y = 0$$

5. Solve by the method of variation of parameters:

   (a) $\frac{d^2y}{dx^2} + a^2y = \sec(ax)$
   (b) $(D^2 - 2D)y = e^x cos(x)$
   (c) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{-x}}{x^2}$
   (d) $\frac{d^2y}{dx^2} + a^2y = \tan(ax)$
   (e) $\frac{d^y}{dx^2} - 3\frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$

6. Solve the following system of differential equations

   (a) $\frac{dx}{dt} - 7x + y = 0, \quad \frac{dy}{dt} - 2x - 5y = 0$
   (b) $\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2cos(t) - 7sin(t), \quad \frac{dx}{dt} - \frac{dy}{dt} + 2x = 4cos(t) - 3sin(t)$
   (c) $\frac{d^2x}{dt^2} - 3x - 4y = 0, \quad \frac{d^2y}{dt^2} + x + y = 0$
   (d) $\frac{dx}{dt} + 2x = t, \quad \frac{dy}{dt} - 2x = \frac{1}{t}$

7. Find the power series solution of the differential equation

$$\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 + 2)y = 0$$

in powers of x (that is, about $x_0 = 0$).

8. Find a power series solution of the initial value problem

$$(x^2 - 1)\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + xy = 0$$

$$y(0) = 4, \quad y'(0) = 6$$